EST II — MATH LEVEL 1 • UNIT 0.01

EST II Math Level 1 Diagnostic & Strategy Drill

PRACTICE WORKSHEET

Core Subject Readiness • Official EST II Curriculum

Name: _____ **Date:** _____ **Score:** _____ / 25

EST II Math Level 1 Diagnostic & Strategy Drill Practice Exercises

1 MCQ

Consider the function $g(x) = 2x + 3$. What is the value of $g(4)$?

A 7

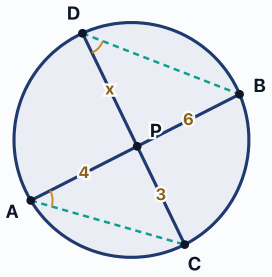
B 8

C 11

D 14

2 MCQ

In the circle shown below, chords AB and CD intersect at point P. If $PA = 4$, $PB = 6$, and $PC = 3$, what is the length of segment $\overline{PD}$?



A 5

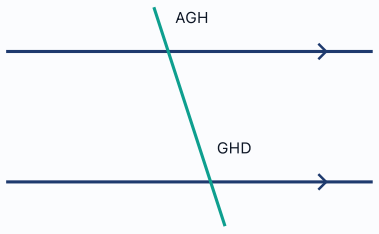
B 7

C 8

D 9

3 MCQ

In the diagram below, line AB is parallel to line CD. Transversal line EF intersects AB at point G and CD at point H. If the measure of angle AGH is 50°, what is the measure of angle GHD?



A 50°

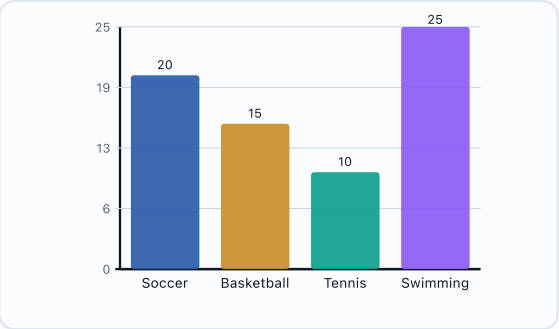
B 130°

C 40°

D 90°

4 MCQ

The bar chart below shows the number of students who chose different sports as their favorite. What is the total number of students surveyed?



A 50

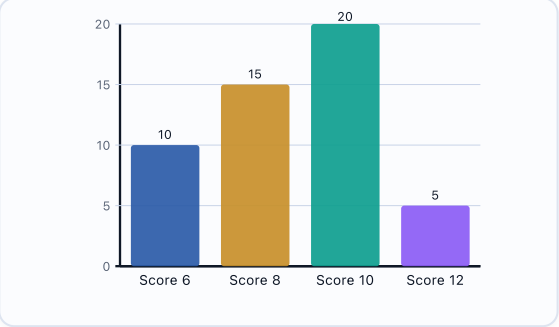
B 60

C 70

D 80

5 MCQ

The bar chart below shows the number of students who achieved different scores on a recent quiz. What is the total number of students who scored 8 or 10?



A 25

B 30

C 40

D 35

6 MCQ

For $\mathbf{a} = \langle 3, 4 \rangle$ and $\mathbf{b} = \langle 4, -3 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 12, -12 \rangle$

B $\langle 9, 0 \rangle$

C $\langle -1, 7 \rangle$

D $\langle 7, 1 \rangle$

7 MCQ

Given the function $f(x) = 2x + 3$. What is the value of $f^{-1}(7)$?

A 1

B 2

C 3

D 4

8 MCQ

For $\mathbf{a} = \langle 4, 7 \rangle$ and $\mathbf{b} = \langle 3, -4 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 9, 2 \rangle$

B $\langle 12, -28 \rangle$

C $\langle 1, 11 \rangle$

D $\langle 7, 3 \rangle$

9 MCQ

For $\mathbf{a} = \langle 5, 10 \rangle$ and $\mathbf{b} = \langle 2, -5 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 7, 5 \rangle$

B $\langle 10, -50 \rangle$

C $\langle 9, 4 \rangle$

D $\langle 3, 15 \rangle$

10 MCQ

For $\mathbf{a} = \langle 6, 13 \rangle$ and $\mathbf{b} = \langle 1, -6 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 7, 7 \rangle$

B $\langle 5, 19 \rangle$

C $\langle 9, 6 \rangle$

D $\langle 6, -78 \rangle$

11 MCQ

For $\mathbf{a} = \langle 7, 16 \rangle$ and $\mathbf{b} = \langle 0, -7 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 7, 23 \rangle$

B $\langle 9, 8 \rangle$

C $\langle 7, 9 \rangle$

D $\langle 0, -112 \rangle$

12 MCQ

For $\mathbf{a} = \langle 8, 19 \rangle$ and $\mathbf{b} = \langle -1, -8 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 9, 27 \rangle$

B $\langle 7, 11 \rangle$

C $\langle -8, -152 \rangle$

D $\langle 9, 10 \rangle$

13 MCQ

For $\mathbf{a} = \langle 9, 22 \rangle$ and $\mathbf{b} = \langle -2, -9 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 9, 12 \rangle$

B $\langle 7, 13 \rangle$

C $\langle 11, 31 \rangle$

D $\langle -18, -198 \rangle$

14 MCQ

For $\mathbf{a} = \langle 10, 25 \rangle$ and $\mathbf{b} = \langle -3, -10 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 9, 14 \rangle$

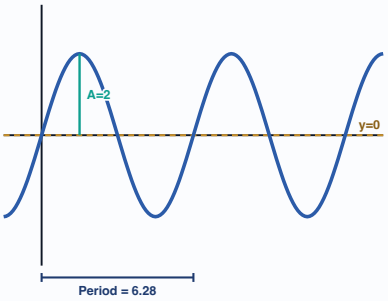
B $\langle -30, -250 \rangle$

C $\langle 7, 15 \rangle$

D $\langle 13, 35 \rangle$

15 MCQ

For $\mathbf{a} = \langle 11, 28 \rangle$ and $\mathbf{b} = \langle -4, -11 \rangle$, what is $\mathbf{a} + \mathbf{b}$?



A $\langle 9, 16 \rangle$

B $\langle -44, -308 \rangle$

C $\langle 7, 17 \rangle$

D $\langle 15, 39 \rangle$

16 MCQ

For $\mathbf{a} = \langle 12, 31 \rangle$ and $\mathbf{b} = \langle -5, -12 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle -60, -372 \rangle$

B $\langle 9, 18 \rangle$

C $\langle 7, 19 \rangle$

D $\langle 17, 43 \rangle$

17 MCQ

Find the angle θ between $\mathbf{u} = \langle 1, 1\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 3 \rangle$.

A 90°

B 60°

C 30°

D 45°

18 MCQ

Calculate the angle θ between $\mathbf{u} = \langle 2, 2\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 4 \rangle$.

A 90°

B 60°

C 30°

D 45°

19 MCQ

Compute the angle θ between $\mathbf{u} = \langle 3, 3\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 5 \rangle$.

A 45°

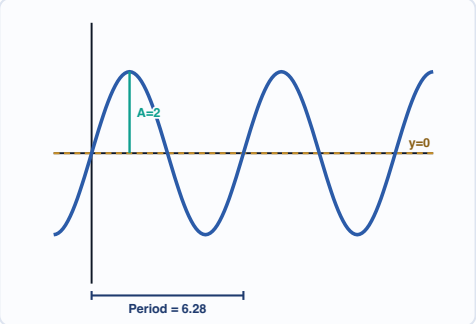
B 30°

C 90°

D 60°

20 MCQ

Deduce the angle θ between $\mathbf{u} = \langle 4, 4\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 6 \rangle$.



A 60°

B 30°

C 90°

D 45°

21 MCQ

Obtain the angle θ between $\mathbf{u} = \langle 5, 5\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 7 \rangle$.

A 60°

B 30°

C 90°

D 45°

22 MCQ

Identify the angle θ between $\mathbf{u} = \langle 6, 6\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 8 \rangle$.

A 45°

B 90°

C 60°

D 30°

23 MCQ

Establish the angle θ between $\mathbf{u} = \langle 7, 7\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 9 \rangle$.

A 90°

B 45°

C 60°

D 30°

24 MCQ

Determine the angle θ between $\mathbf{u} = \langle 8, 8\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 10 \rangle$.

A 60°

B 30°

C 90°

D 45°

25 MCQ

Calculate the angle θ between $\mathbf{u} = \langle 9, 9\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 11 \rangle$.

A 60°

B 90°

C 30°

D 45°